

YEAR 11 Mathematics Extension 1 – 2005 Yearly Examination (Term 4)

Question 1.

Marks

- (a) An arc of length 10 cm subtends an angle of θ radians at the centre of a circle of radius 5 cm. **2**
Find the value of θ correct to the nearest minute.
- (b) Find the equation of the curve passing through the point $(1, 3)$ with a gradient function of $(x+1)(x-5)$. **3**
- (c) Find the primitive function of
- (i) $2 \sin 4x$. **2**
- (ii) $\frac{x+3}{x^2+5}$. **3**

Question 2.

[START A NEW PAGE]

- (a) The point A lies on the line $3x+2y=24$. A line, perpendicular to the x -axis, is drawn through point A and meets the x -axis at B .
- (i) If B is $(a, 0)$, find the coordinates of A in terms of a . **1**
- (ii) The triangle bounded by the lines AB , $3x+2y=24$ and the x -axis has an area of 27 square units, find the coordinates of A . **3**
- (b) (i) Given that A is $(-3, 7)$, B is $(-2, 12)$, C is (x, y) and D is $(2, 8)$. **1**
Find the coordinates of C if $ABCD$ is a rhombus.
- (ii) Hence, find the area of $ABCD$. **3**
- (c) Find the integral of $4e^x + \sqrt{x}$. **2**

Question 3. [START A NEW PAGE]

- (a) (i) Express $\sqrt{3} \sin \theta + \cos \theta$ in the form $A \sin(\theta + \alpha)$, where $A > 0$ and $0 < \alpha < 2\pi$. **2**
- (ii) Find the minimum value of $\sqrt{3} \sin \theta + \cos \theta$ and determine when this minimum first occurs for $\theta \geq 0$. **2**
- (iii) Neatly sketch $y = \sqrt{3} \sin \theta + \cos \theta$, for $0 \leq \theta \leq 2\pi$, clearly showing all important features. **3**
- (b) Using the t -results, solve $\cos A + \sqrt{3} \sin A = -1$, for $0 \leq A \leq 2\pi$ **3**

Question 4. [START A NEW PAGE]

Marks

- (a) Find the exact area bounded by $y = \cos^{-1} x$, the x -axis and the lines at $x = 0$ and $x = \frac{1}{\sqrt{2}}$. **3**
- (b) Prove that $\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{4}$. **3**
- (c) Find $\int_1^3 \frac{(x-1)}{(x+1)^3} dx$, using the substitution $u = x + 1$. **4**

Question 5. [START A NEW PAGE]

- (a) If $A = \sin^{-1}\left(\frac{5}{13}\right)$, find the value of $\sin 2A$. **2**
- (b) Given $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$, **4**
and $p + q = 2\sqrt{3}$.
Find the angle between the chord PQ and the axis of the parabola.
- (c) P is a point on $x^2 = 12y$ and O is the Origin. Q is the foot of the perpendicular **4**
from the focus, S , of the parabola to OP .
Show that the locus of Q is given by $x^2 + y^2 - 3y = 0$.

Question 6. [START A NEW PAGE]

- (a) Find the inverse function of $y = 3 + \ln x$. **1**
- (b) Differentiate $y = \frac{1}{2} \tan^{-1} x$ with respect to x . **1**
- (c) Neatly sketch $y = \frac{1}{2} \tan^{-1} x$ and its derivative. **3**
- (d) Find the domain and range of $y = \sin^{-1}\left[\frac{1}{2(1+x^2)}\right]$. **3**
- (e) Neatly sketch $y = \sin^{-1}\left[\frac{1}{2(1+x^2)}\right]$. **2**

THE END

T4 - Bunt Solutions - 2005

QUESTION 1

a) $l = re$

$\theta = \frac{10}{5} = 2$

$\theta = 2 \times \frac{180}{\pi} = 114^\circ 35'$

b) $m = (x+1)(x-5)$

$= x^2 - 4x - 5$

$\therefore y = \frac{1}{3}x^3 - 2x^2 - 5x + C$

sub in (1, 3)

$3 = \frac{1}{3} - 2 - 5 + C$

$\therefore C = -12/3$

$\therefore y = \frac{1}{3}x^3 - 2x^2 - 5x - \frac{5}{3}$

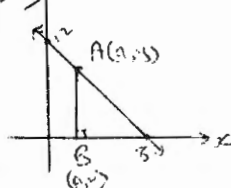
c) i) $-\frac{1}{2} \cos 4x + C$

ii) $\int (\frac{x}{x^2+5} + \frac{3}{x^2+5}) dx$

$= \frac{1}{2} \ln(x^2+5) + \frac{3}{\sqrt{5}} \tan^{-1}(\frac{x}{\sqrt{5}}) + C$

QUESTION 2

a) $y = 12 - 3/2x$



A lies on

$3x + 2y = 24$

$\therefore 3a + 2y = 24$

$2y = 24 - 3a$

$y = 12 - 3/2a$

$\therefore A(a, 12 - 3/2a)$

ii) $A = \frac{1}{2}bh$

$27 = \frac{1}{2}(8-a)(12-3/2a)$

$54 = 48 - 12a - 12a + \frac{3}{2}a^2$

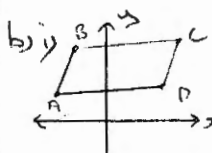
$0 = \frac{3}{2}a^2 - 24a + 42$

$0 = a^2 - 16a + 28$

$0 = (a-14)(a-2)$

$\therefore a = 14$ or $a = 2$

$\therefore A(2, 9)$ or $(14, -9)$



ii) $A = \frac{1}{2}xy$
 $= \frac{1}{2}(\sqrt{16+16})(\sqrt{36+36})$
 $= 24 \text{ units}^2$

c) $\int (4e^{2x} + \sqrt{x}) dx$

$= 2e^{2x} + \frac{2}{3}x^{3/2} + C$

QUESTION 3

a) i) $\sqrt{3} \sin \theta + \cos \theta \equiv A \sin(\theta + \alpha)$
 $= A \sin \theta \cos \alpha + A \cos \theta \sin \alpha$

$\frac{\sqrt{3}}{A} = \cos \alpha$

$\frac{1}{A} = \sin \alpha$

$\frac{3}{A^2} + \frac{1}{A^2} = 1$

$A^2 = 4$

$A = 2$ as $A > 0$

$\tan^{-1} \frac{1/\sqrt{3}}{3/2} = \alpha$

$\alpha = \pi/6$

$\therefore \sqrt{3} \sin \theta + \cos \theta = 2 \sin(\theta + \pi/6)$

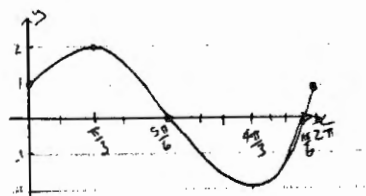
ii) Minimum value is -2

first occurs when $\sin(\theta + \pi/6) = -1$

$\theta + \pi/6 = 3\pi/2$

$\theta = \frac{4\pi}{3}$

iii) $y = \sqrt{3} \sin \theta + \cos \theta$
 $\therefore y = 2 \sin(\theta + \pi/6)$



① for endpoints

① intercepts

① range/shape

b) $\cos A + \sqrt{3} \sin A = -1$

check for $A = \pi$

$\cos \pi + \sqrt{3} \sin \pi = -1$

$-1 + 0 = -1$

true

$\therefore \pi = A$ is a soln.

$1 - t^2 + 2\sqrt{3}t = -1 - t^2$

$2\sqrt{3}t = -2$

$t = -\frac{1}{\sqrt{3}}$

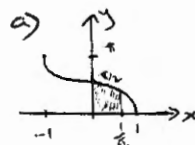
$\therefore \tan \frac{A}{2} = -\frac{1}{\sqrt{3}}$

$\frac{A}{2} = \frac{5\pi}{6}$

$\therefore A = \frac{5\pi}{3}$

$\therefore$ soln. set is $A = \frac{5\pi}{3}$ or π

QUESTION 4



Area = $\int_{-\pi/2}^{\pi/2} \cos x dx$

$= \int_{-\pi/2}^{\pi/2} \cos y dy + (\frac{1}{\sqrt{2}} + \frac{\pi}{4})$

$= [\sin y]_{-\pi/2}^{\pi/2} + \frac{\pi}{4\sqrt{2}}$

$= \sin \pi/2 - \sin -\pi/2 + \frac{\pi}{4\sqrt{2}}$

$= (1 - \frac{1}{\sqrt{2}} + \frac{\pi}{4\sqrt{2}}) \text{ units}^2$

b) $\tan \frac{\pi}{4} = \tan(A+B)$
 where $A = \tan^{-1}(\frac{1}{3})$ & $B = \tan^{-1}(\frac{1}{2})$
 $\tan A = \frac{1}{3} \therefore 0 < A < \frac{\pi}{4}$
 $\tan B = \frac{1}{2} \therefore 0 < B < \frac{\pi}{4}$ (1)

RHS. = $\tan(A+B)$
 $= \frac{\tan A + \tan B}{1 - \tan A \tan B}$
 $= \frac{\frac{1}{3} + \frac{1}{2}}{1 - \frac{1}{3} \times \frac{1}{2}} = \frac{\frac{5}{6}}{\frac{5}{6}}$
 $= 1$

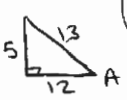
$\therefore \tan \frac{\pi}{4} = 1$ (1)
 $\tan^{-1} 1 = \frac{\pi}{4}$
 $\therefore \tan^{-1}(\frac{1}{3}) + \tan^{-1}(\frac{1}{2}) = \frac{\pi}{4}$

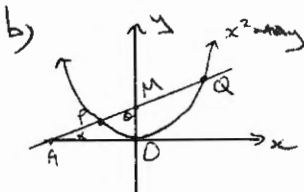
c) $\int_1^3 \frac{x-1}{(x+1)^3} dx$ $u = x+1$
 $\frac{du}{dx} = 1$

when $x=1, u=2$
 $x=3, u=4$

$\therefore \int_2^4 \frac{u-2}{u^3} du = \int_2^4 (u^{-2} - 2u^{-3}) du$
 $= \left[-\frac{1}{u} + \frac{1}{2} u^{-2} \right]_2^4$
 $= \left(-\frac{1}{4} + \frac{1}{256} \right) - \left(-\frac{1}{2} + \frac{1}{32} \right)$
 $= -\frac{63}{256} + \frac{15}{32}$
 $= \frac{57}{256}$

QUESTION 5

a) $A = \sin^{-1}(\frac{5}{13})$
 $\sin A = \frac{5}{13}$ for $-\frac{\pi}{2} \leq A \leq \frac{\pi}{2}$
 but $\sin A > 0 \therefore 0 \leq A \leq \frac{\pi}{2}$
 (1)
 $\sin 2A = 2 \sin A \cos A$
 $= 2 \left(\frac{5}{13} \right) \left(\frac{12}{13} \right)$
 $= \frac{120}{169}$ (1)

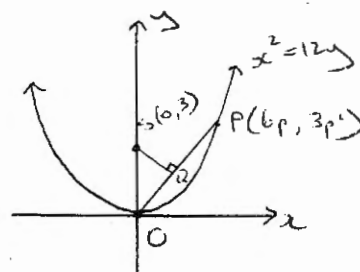


$m = \tan \alpha$
 $\therefore \frac{a(q^2 - p^2)}{2a(q-p)} = \tan \alpha$ (1)
 $\frac{p+q}{2} = \tan \alpha$
 $\frac{2}{2} = \tan \alpha$ (as $p+q=2$)
 $1 = \tan \alpha$
 $\therefore \alpha = \frac{\pi}{4}$ (1)

(1) $\angle AOM = \frac{\pi}{2}$ (y-axis is perpendicular to x-axis)

(1) $\therefore \theta = \frac{\pi}{4}$ (angle sum of $\triangle AOM$ is π)

$\therefore PQ$ makes constant angle of $\frac{\pi}{4}$ with the axis of parabola $x^2 = 4ay$.



eqn. of PO is: $y-0 = \frac{3p^2}{6p}(x-0)$
 $y = \frac{p}{2}x$... (1)

eqn. of SQ is: $y-3 = -\frac{3}{p}(x-6p)$
 $y = -\frac{3}{p}x + 3$... (2)

solve (1) & (2) to get Q

$\frac{p}{2}x = -\frac{3}{p}x + 3$

$x \left(\frac{p}{2} + \frac{3}{p} \right) = 3$

$x \left(\frac{p^2+4}{2p} \right) = 3$

$\therefore x = \frac{6p}{p^2+4}$ (1)

$\therefore y = \frac{p}{2} \left(\frac{6p}{p^2+4} \right)$

$= \frac{3p^2}{p^2+4}$

$\therefore Q \left(\frac{6p}{p^2+4}, \frac{3p^2}{p^2+4} \right)$ (1)

Locus of Q is $x^2 + y^2 - 3y = 0$... (2)

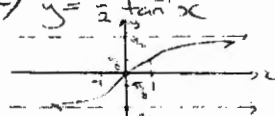
LHS. = $x^2 + y^2 - 3y$
 $= \frac{36p^2}{(p^2+4)^2} + \frac{9p^4}{(p^2+4)^2} - \frac{9p^2}{p^2+4}$
 $= \frac{36p^2 + 9p^4 - 9p^2(p^2+4)}{(p^2+4)^2}$ (1)
 $= 0$

$\therefore$ Locus of Q is the circle $x^2 + y^2 - 3y = 0$ (1) with centre $(0, \frac{3}{2})$, radius $\frac{3}{2}$.

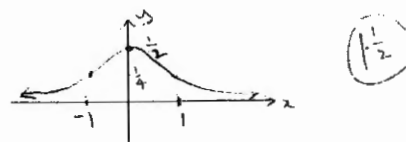
QUESTION 6

a) $y = 3 + \ln x$
 inverse is $x = 3 + \ln y$
 $x - 3 = \ln y$
 $\therefore y = e^{x-3}$ (1)

b) $y = \frac{1}{2} \tan^{-1} x$
 $\frac{dy}{dx} = \frac{1}{2(1+x^2)}$ (1)

c) $y = \frac{1}{2} \tan^{-1} x$
 (1/2)

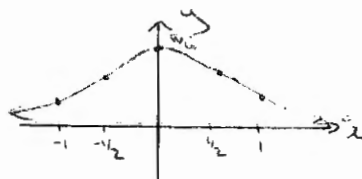
$y = \frac{1}{2(1+x^2)}$



d) Domain: $x \in \mathbb{R}$ (1)
 Range: $0 < y \leq \frac{\pi}{6}$ (2)

Locus of Q is $x^2 + y^2 - 3y = 0$... (2)

LHS. = $x^2 + y^2 - 3y$
 $= \frac{36p^2}{(p^2+4)^2} + \frac{9p^4}{(p^2+4)^2} - \frac{9p^2}{p^2+4}$
 $= \frac{36p^2 + 9p^4 - 9p^2(p^2+4)}{(p^2+4)^2}$ (1)
 $= 0$



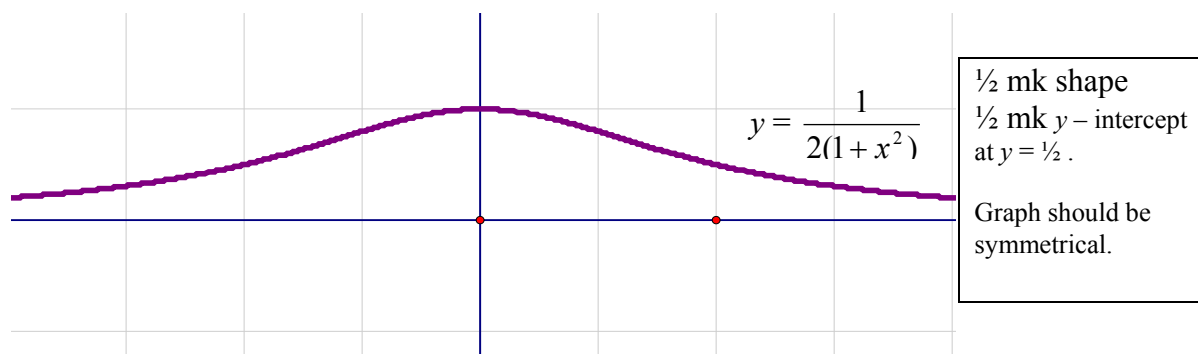
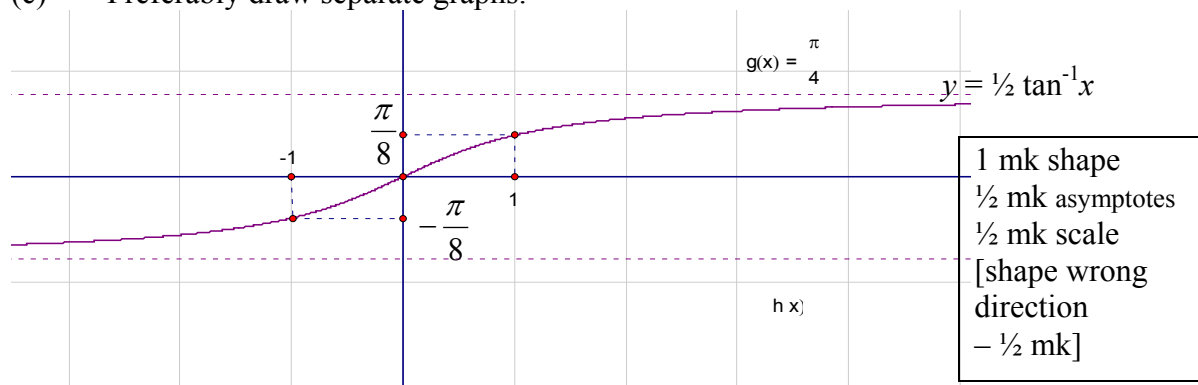
(1) for shape $(\frac{1}{2}, 0)$ to $(\frac{1}{2}, \frac{\pi}{6})$
 (1/2) (1/6) complete

Term 4 Ext 1 Maths QUESTION 6 Marking Guidelines (L. Kim)

- (a) $y = 3 + \ln x \rightarrow$ swap x and y
 $\therefore x - 3 = \ln y \rightarrow y = e^{x-3}$ **1 mk** No half marks awarded.

- (b) $y = \frac{1}{2} \tan^{-1} x \quad \therefore \frac{dy}{dx} = \frac{1}{2(1+x^2)}$ **1 mk** No half marks

- (c) Preferably draw separate graphs.



- (d) From part (c) second graph, the domain of $y = \frac{1}{2(1+x^2)}$ is $\{x : x \in \mathbb{R}\}$.

Also the range is $0 < y \leq \frac{1}{2} \quad \therefore \sin^{-1}(0) < \sin^{-1}\left(\frac{1}{2(1+x^2)}\right) \leq \sin^{-1}\left(\frac{1}{2}\right)$

$\therefore \text{FOR } y = \sin^{-1} \frac{1}{2(1+x^2)}$

DOMAIN is $\{x : x \in \mathbb{R}\}$ **1mk** and the RANGE is $0 < y \leq \frac{\pi}{6}$. **2mk- 1/2 mk each**

- (e)

